EXPERIMENT 13: Transfer Files from One Virtual Machine to Another Virtual Machine

1. Aim

To establish communication between two virtual machines and transfer a file using SSH and SCP.

2. Procedure

1. Install VirtualBox or VMware Workstation on the host computer.
2. Create two virtual machines named VM1 and VM2.
3. Install Linux on both virtual machines.
4. Configure both virtual machines to use the same reachable network, such as a host-only network or bridged network.
5. Start both virtual machines and log in.
6. Install the OpenSSH server on both virtual machines.
7. Find the IP address of VM1 using hostname -I.
8. Find the IP address of VM2 using hostname -I.
9. From VM1, test connectivity to VM2 using ping and its IP address.
10. Create a sample file in VM1 using echo and redirect the text into sample.txt.
11. Use the scp command on VM1 to copy sample.txt to the home directory of the user on VM2.
12. Enter the VM2 user's password when prompted, if password authentication is enabled.
13. Log in to VM2 and use ls to verify that sample.txt has arrived.
14. Display the file contents using cat sample.txt.

3. Program

Install SSH and identify addresses:

```
sudo apt update
sudo apt install openssh-server -y
hostname -I
ping <VM2-IP>
```

Create and transfer a file from VM1:

```
echo "File Transfer Successful" > sample.txt
scp sample.txt username@<VM2-IP>:/home/username/
```

Verify on VM2:

```
ls
cat sample.txt
```

4. Output of Program

```
VM1:
$ scp sample.txt user@192.168.56.102:/home/user/
sample.txt 100% transferred

VM2:
$ ls
sample.txt

$ cat sample.txt
File Transfer Successful
```

Note: Replace the username and example IP address with the actual VM2 username and IP address. Both virtual machines must be able to reach each other over the selected network.

5. Result

Thus, two virtual machines were connected through a network, and a file was transferred from VM1 to VM2 using SCP.

EXPERIMENT 13: Transfer Files from One Virtual Machine to Another Virtual Machine

1. Aim

To establish network communication between two Linux virtual machines and transfer files securely from one virtual machine to another using SSH and the SCP command.

2. Procedure

1. Install Oracle VirtualBox or VMware Workstation on the host computer. 2. Create two virtual machines named VM1 and VM2 using the virtualization software. 3. Install a Linux operating system on both virtual machines. 4. Start both virtual machines and log in using their respective user accounts. 5. Configure both virtual machines to use the same reachable network, such as a host-only network or bridged network. 6. Verify the network configuration and ensure that both virtual machines can communicate with each other. 7. Open the terminal in VM1 and install the OpenSSH server package using the package manager. 8. Repeat the OpenSSH installation process in VM2. 9. Execute hostname -I in VM1 to identify its IP address. 10. Execute hostname -I in VM2 to identify its IP address. 11. From VM1, use the ping command with VM2's IP address to test network connectivity. 12. If the connection is successful, create a sample text file named sample.txt in VM1. 13. Enter sample data into the file using a text editor or the echo command. 14. Execute the SCP command from VM1 to copy the file to the home directory of the user in VM2. 15. Enter the destination user's password if password authentication is enabled. 16. Wait until the file transfer process is completed successfully. 17. Log in to VM2 and navigate to the destination directory. 18. Execute the ls command to verify that sample.txt has been transferred. 19. Execute cat sample.txt to display the contents of the transferred file. 20. Compare the file contents with the original file in VM1 to confirm successful transfer.

4. Output of Program

VM1 – File Transfer Command: $ scp sample.txt user@192.168.56.102:/home/user/ sample.txt 100% transferred VM2 – File Verification: $ ls sample.txt Displaying File Contents: $ cat sample.txt File Transfer Successful Explanation: SCP stands for Secure Copy Protocol. It uses SSH to transfer files securely between networked computers. In this experiment, VM1 acts as the source machine and VM2 acts as the destination machine.

5. Result

Thus, two Linux virtual machines were successfully configured on the same network. Network connectivity was established using SSH, and a sample file was securely transferred from VM1 to VM2 using the SCP command.